

HANNING LU

BSc Computer Science Candidate, University of Leeds

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RESEARCH PROFILE

Computer Science undergraduate working on machine learning systems and efficient large model inference, with experience in KV cache optimization, SGLang based serving, GPU profiling, model compression, and distributed generative model serving. Interested in the interaction between model level information and runtime system state, especially how cached model state can be represented, managed, and reused efficiently across large scale inference systems.

EDUCATION

University of Leeds

Sep 2024 – Jun 2027 (Expected)

Bachelor of Science in Computer Science | Leeds, United Kingdom

GPA 3.8/4.0, expected to graduate with a 1:1 degree.

RESEARCH INTERESTS

Focus areas: Machine learning systems; LLM serving; long context inference; KV cache management; hierarchical memory and storage; distributed inference; model compression; GPU systems optimization; model and runtime co design.

Current question: How can large model serving systems jointly use model level signals about the importance and fidelity of cached state together with runtime signals about reuse, memory pressure, data movement, and recomputation cost to make efficient KV management decisions across the serving hierarchy?

RESEARCH & TECHNICAL EXPERIENCE

Efficient LLM Inference and Model Optimization

Jun 2024 – Jun 2025

School of Computer Science, Peking University | Research Assistant, Prof. Tong Yang's Group

- Developed a dynamic KV-cache compression strategy that models attention-score trajectories with ARIMA and predicts head-specific token importance for adaptive cache retention and eviction; on Llama-2-7B and LongBench, reported 70% compression with approximately 1% performance loss.
- Explored vector-database retrieval and controlled generation for RAG-based advertising integration, and conducted Transformer quantization and structured/non-structured pruning experiments for lower-memory, lower-latency inference.

Video Generation Inference Infrastructure

Jun 2026 – Present

Xingjie Intelligence | Video Generation Infrastructure Engineer Intern

- Develop SGLang-based serving infrastructure for diffusion models, optimizing execution through tensor, pipeline, and sequence parallelism, KV caching, and block-wise autoregressive generation.
- Profile GPU bottlenecks with Nsight Systems, Nsight Compute, and PyTorch Profiler; explore FP8, weight-only, and activation quantization together with CUDA Graph, kernel fusion, continuous batching, and streaming inference, using FPS, chunk latency, TTFT, memory use, and scaling efficiency as evaluation metrics.

LLM Agents for Compiler Optimization and LLVM IR Decompilation

May 2025 – Jun 2026

Distributed Computing Group, University of Leeds | Research Assistant

- Built a closed-loop LLM-agent workflow linking performance profiling, optimization decisions, compilation feedback, and strategy updates; integrated VTune/perf signals, RAG-based retrieval of optimization knowledge, and automated LLVM pass tuning.
- Evaluated the optimization workflow on CBench and PolyBench, reporting a $1.55\times$ average speedup over -O3 on CBench and up to $16.7\times$ on selected compute-intensive PolyBench benchmarks.
- Designed an LLM-assisted LLVM IR decompilation pipeline using chain-of-thought prompting, iterative feedback, and model self-correction; reported over 90% accuracy after iterative optimization on obfuscated IR tests.

Reinforcement Learning & Graph Representation Learning for Trustworthy AI

Oct 2025 – Jul 2026

National Engineering Laboratory for Big Data Analysis and Applications, Peking University | Research Assistant, Bin Chong's Team

- VeriBot: Developed an evidence-grounded Qwen3-0.6B verifier post-trained with SFT and Active-GRPO, designing gated rewards for grounded, sufficient, and non-redundant evidence reasoning; integrated the independently trained LLM view with graph predictions for conflict-aware verification and selective abstention.
- SAHG: Designed a sector-anisotropic hyperbolic graph model with learnable direction-dependent curvature, angular sector prototypes, and decoupled node/neighborhood channels to model heterogeneous social-graph structure and mitigate heterophilic message-passing contamination; achieved state-of-the-art ACC/F1 across three evaluated benchmarks.

- Designed and optimized multi-agent coordination code and configurable task-scheduling workflows to support more accurate data collection, stable execution, fault tolerance, and recovery in complex browser-based tasks.

PUBLICATIONS & MANUSCRIPTS

- **H. Lu**, C. Xia. *Agent-Agnostic End-to-End C/C++ Application Performance Optimization*. Proceedings of the 40th ACM International Conference on Supercomputing Workshops, 2026. DOI: 10.1145/3774895.3812199.
- W. Zhong*, **H. Lu***, Z. Yao, S. Zhang, C. Xia. *Revision Propensity Is Not Revision Reliability: Ground-Truth-Conditioned Evaluation of Prompt Framing in LLM Cascades*. Manuscript, 2026. *Co-first authors.
- **H. Lu**, Y. Yang, J. Su, Y. Liu, Z. Yao, Y. Li, T. Liang, Z. Zhang, R. Ran, K. Xu, B. Chong. *SAHG: Sector-Anisotropic Hyperbolic Graph Model for Social Bot Detection*. arXiv preprint, 2026. arXiv:2605.30166.
- **H. Lu**, K. Hu, Y. Yang, Y. Liu, Z. Yao, Y. Li, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, B. Chong. *VeriBot: Evidence-Routed Social Bot Detection with Auditable Decision Chains*. Manuscript under review, 2026.
- Z. Yao, **H. Lu**. *HCGM: Hyperbolic Code Graph Modeling for Repository-Level Software Intelligence*. Manuscript under review, 2026.
- K. Hu, Y. Yang, **H. Lu**, Y. Liu, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, H. Peng, P. S. Yu, B. Chong. *BotRoute: Dual-Branch Selective Routing for Social Bot Detection*. Manuscript under review, 2026.
- Z. Yao, Y. Tang, **H. Lu**, Y. Li, M. Yang. *Hierarchy-Aware Sparse Autoencoders via Activation-Cone Routing*. Manuscript under review, 2026.
- H. Jia, Y. Liu, B. Chong, Y. Yang, Y. Chen, J. Liang, Q. Li, **H. Lu**, K. Xu, H. Zheng, C. Zhang, H. Peng, Philip S. Yu. *FinHarness: An Inline Lifecycle Safety Harness for Finance LLM Agents*. arXiv preprint, 2026. arXiv:2605.27333.
- Y. Wang, T. Yang, X. Liang, G. Wang, **H. Lu**, Z. Xu, Y. Li, W. Li. *Art and Science of Quantizing Large-Scale Models: A Comprehensive Overview*. arXiv preprint, 2024. arXiv:2409.11650.

HONORS & AWARDS

- Gold Medal, China Collegiate Programming Contest (CCPC), Zhengzhou Invitational — Team Captain 2024
- Silver Medal, 48th International Collegiate Programming Contest (ICPC), Xi'an Invitational — Team Captain

RESEARCH METHODS & SKILLS

Programming and systems: Python; C/C++; LLVM; SGLang; CUDA Graph; tensor, pipeline, and sequence parallelism; VTune; perf; Nsight Systems; Nsight Compute; PyTorch Profiler.

Machine learning and generative AI: PyTorch; LLM agents; self-evolving/self-improving agent systems; RAG; Llama; Wan diffusion models; KV-cache optimization; model quantization and pruning; hyperbolic graph neural networks; knowledge graphs and graph representation learning.

Research methods: Benchmark design and evaluation; ablation studies; performance profiling; time-series modeling; graph representation learning; knowledge graph construction and reasoning; reproducible empirical comparison.